Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

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Research Interests

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

Education

2021 – ongoing **Ph.D., Electrical and Electronics Engineering**
Northfield Institute of Science and Technology
Thesis: Lightweight deep networks for real-time enhancement of aerial imagery
Supervisor: Prof. R. Iyer (fictional)

2010 **M.Tech., Avionics**
Coastal Institute of Aeronautical Engineering, Chennai
Grade: CGPA 8.6/10
Thesis: Fault-tolerant data acquisition for airborne test instrumentation

2004 **B.E., Electronics and Communication Engineering**
Northfield Institute of Science and Technology, Northfield University, Pune
Grade: First Class with Distinction

Appointments

2016 – Present **Scientist E**
Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi
Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop 2019 – Present
Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

- Supervise technicians and plan preventive maintenance for workshop test equipment.
- Introduced a job-card and spares-inventory system for instrument repairs and calibration work.

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Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** 2021 – Present

2004 – 2016 **Senior Design Engineer**

Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru

Integration and qualification testing of airborne electronics and UAV payloads.

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.
- Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.
- Planned environmental qualification (MIL-STD-810 vibration, temperature and humidity) for line-replaceable units and documented the test results.
- Mentored graduate engineer trainees on avionics bus protocols and test documentation.

2002 – 2004 **Graduate Engineer**

Test Systems Group, Meridian Electronics Limited, Bengaluru

Built automated test equipment for avionics line-replaceable units.

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

Key Projects

2021 – Present **Edge super-resolution pipeline for UAV imagery** *(Ongoing)*

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

2018 – 2019 **Radiosonde ground-station acceptance test protocol**

Coordinator

Station acceptance testing varied between sites.

- A common acceptance protocol and checklist adopted for new ground-receiving systems.

2020 – 2022 **IoT monitoring of laboratory infrastructure**

Project lead

Instrument rooms lacked continuous environmental and power-quality monitoring.

Technologies: ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

- Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.
- Continuous monitoring with automated alerts replaced manual twice-daily readings.

Publications

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

Manuscripts Under Review / In Preparation

[1] **A. Sharma**, M. Khan and R. Iyer, "Detection-aware quality assessment of super-resolved drone images," under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] **A. Sharma**, "Simulation study of balloon-released gliding sondes," in preparation. *[In preparation]*

Research Grants & Funding

2023 – 2025 **Seed grant: edge-AI enhancement of aerial imagery** *(Completed)*
Principal investigator, Example Science Foundation (fictional)
Two-year seed project on lightweight super-resolution networks for embedded GPUs.
Reference: ESF/SEED/2023/017

Talks & Presentations

2024 **Invited talk: Edge AI for atmospheric observation**
Example Symposium on Applied AI

Awards & Honours

2023 **Best Paper Award**
Example Conference on Instrumentation (ECI 2023)

Professional Service

2022 – Present **Reviewer, Example Journal of Atmospheric Instrumentation**
Example Journal of Atmospheric Instrumentation

2024 – 2025 **Co-supervisor, two M.Tech. dissertations in embedded vision**
Example Institute of Technology (fictional)

Professional Memberships

2010 – Present **Member, Institution of Example Engineers**
Institution of Example Engineers

Service & Leadership

2022 – Present **Supervisor, M.Tech. interns in applied machine learning**
Kestrel Atmospheric Instrumentation Laboratory
Supervise summer and dissertation interns on edge-AI projects.

2023 – 2024 **Chair, Technical Evaluation Committee for radiosonde procurement**
Kestrel Atmospheric Instrumentation Laboratory
Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

Certifications & Training

2022 **Deep Learning for Edge Devices (40-hour course)**
Example Academy of Applied Computing

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2019 **Public Procurement and GFR 2017 (5-day programme)**
Example Institute of Public Administration

Technical Skills

Avionics: MIL-STD-1553 / ARINC 429, Avionics integration
Scientific instrumentation: Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems
Embedded systems: Embedded C, ESP32 / STM32 microcontrollers
Computer vision: Image super-resolution, Object detection
Drones & UAV systems: UAV payload integration
Edge AI: NVIDIA Jetson deployment
AI / machine learning: Deep learning, PyTorch
Procurement & administration: Technical procurement (GeM, GFR)
IoT: MQTT and Node-RED, InfluxDB and Grafana
Model optimisation: TensorRT and ONNX, Pruning and knowledge distillation
Software & tools: Python, Docker and Linux
Technical leadership: Team leadership
GPU computing: CUDA / GPU computing

Other Activities

2020 – 2021 **Lead author, upper-air instrument maintenance manual (internal)**
Kestrel Atmospheric Instrumentation Laboratory

Languages

English, Hindi

Referees

Dr. R. Iyer (fictional)
Professor of Electronics, Example Institute of Technology
Research collaborator and co-author
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